The empirical recurrence for OEIS A237071: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A237071 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 640$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237071 is “Number of (n+1)X(3+1) 0..3 arrays with the upper median of every 2X2 subblock equal.”. It has offset 1 and begins

$$15882, 1046822, 53940782, 3263989086, 195683299844, 12159520038142, 763699683972418, 48556387347813958, .$$

The entry states:

> Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:48:08, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 640$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most* $S' =$ *640.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1280$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{62} c_i \, a(n-i),$$

c_1	114	c_{17}	−7233761941528462563844	c_{33}	−61485524587162550515
c_2	−163	c_{18}	−4605394619111953114730273	c_{34}	−22338754866379359505
c_3	−346722	c_{19}	6717609462978417472077162	c_{35}	1077632621431578821919
c_4	5988183	c_{20}	412835648195937717826932071	c_{36}	3078956965999425670334
c_5	435280312	c_{21}	−986443030543893918279775360	c_{37}	−148602656767379476744
c_6	−9816206042	c_{22}	−28593236054612804349233691681	c_{38}	−339488887593856137362
c_7	−296067869512	c_{23}	88285565834871804496517774996	c_{39}	16083647369487084234299
c_8	7712693710548	c_{24}	1547784066952260886041503938725	c_{40}	29903298000104665905320
c_9	120857038182028	c_{25}	−5604618360179369595884515343406	c_{41}	−1359852427800589393909
c_{10}	−3641506878305292	c_{26}	−66041842192536404380975388966449	c_{42}	−2098304621642605227731
c_{11}	−30859803019498898	c_{27}	266292778266321948793737965833664	c_{43}	89179801689429720985725
c_{12}	1135745702063236569	c_{28}	2234965738231263441110737097653692	c_{44}	116704074978164006647938
c_{13}	4866767027249268624	c_{29}	−9715931931703399669213792037631104	c_{45}	−44918704004616321493262
c_{14}	−247398750401791577401	c_{30}	−60245403501327020981332914765890624	c_{46}	−51012689926124439224286
c_{15}	−400248385669338194212	c_{31}	275993327704776383398382503579697664	c_{47}	171457209100803469804535
c_{16}	39092165381593028611836	c_{32}	1297088062931736755450489990749869440	c_{48}	172897358757723388115215

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A237071.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.